THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 7 cm	The third side lies between the difference and the sum of the other two.
2	**(c)** 30	Add the angles: $(3+k)x = 180$.
3	**(c)** 30	Exterior angle = sum of interior opposite angles.
4	**(d)** QR	Use the idea of corresponding parts.
5	**(a)** $\angle A$	The included angle lies between the two given sides.
6	**(b)** $50°, 60°, 70°$	The three angles must add up to $180°$.
7	**(b)** $\angle B = \angle C$	Equal angles and equal sides go together in an isosceles triangle.
8	**(b)** not necessarily congruent	AAA does not guarantee congruence.
9	**(b)** 10 cm	The third side lies between the difference and the sum of the other two.
10	**(b)** 20	Add the angles: $(3+k)x = 180$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Third side $> 6 - 10$ in absolute value $= 4$; Third side $< 6 + 10 = 16$ (1); Least whole number 5 cm; Greatest whole number 15 cm (1).	1+1
12	$6x + 18 = 180$; $x = 27$ (1); Angles $64°$, $77°$, $39°$ (1).	1+1
13	Exterior angle $= \angle A + \angle B$; $5x - 10 = 3x + 50$, so $2x = 60$ (1); $x = 30$; $\angle ACD = 140°$ (1).	1+1
14	Third side $> 4 - 13$ in absolute value $= 9$; Third side $< 4 + 13 = 17$ (1); Least whole number 10 cm; Greatest whole number 16 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = \angle C$ (equal sides); $(x + 10) + 2(2x - 5) = 180$, so $5x = 180$ (1); $x = 36$ (1); $\angle A = 46°$, $\angle B = \angle C = 67°$ (1).	1+1+1
16	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 9$ cm (1); Perimeter of PQR $= 5 + 7 + 9$ (1); $= 21$ cm (1).	1+1+1
17	$\angle A + \angle B = 120°$; One part $= 120 \div 4 = 30°$ (1); $\angle A = 30°$, $\angle B = 90°$ (1); $\angle C = 180° - 120° = 60°$ (1).	1+1+1
	OR: $\angle B = \angle C$ (equal sides); $(x + 20) + 2(2x - 15) = 180$, so $5x = 190$ (1); $x = 38$ (1); $\angle A = 58°$, $\angle B = \angle C = 61°$ (1).	1+1+1
18	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 12$ cm (1); Perimeter of PQR $= 6 + 9 + 12$ (1); $= 27$ cm (1).	1+1+1
	OR: $\angle A + \angle B = 140°$; One part $= 140 \div 7 = 20°$ (1); $\angle A = 40°$, $\angle B = 100°$ (1); $\angle C = 180° - 140° = 40°$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$4x + 20 = 180$, so $x = 40$ (1); Angles $50°$, $30°$, $100°$ (1); Classification: obtuse-angled (1); Exterior angles $130°$, $150°$, $80°$ (1); Their sum $= 360° = 360°$ (1).	1+1+1+1+1
	OR: (i) Between 5 cm and 19 cm (1); (ii) Whole numbers 6 to 18: 13 values (1); (iii) Smallest third side 6: perimeter 25 cm (1); Largest third side 18: perimeter 37 cm (1); State the answers with units (1).	1+1+1+1+1
20	$4x + 20 = 180$, so $x = 40$ (1); Angles $60°$, $30°$, $90°$ (1); Classification: right-angled (1); Exterior angles $120°$, $150°$, $90°$ (1); Their sum $= 360° = 360°$ (1).	1+1+1+1+1
	OR: (i) Between 4 cm and 14 cm (1); (ii) Whole numbers 5 to 13: 9 values (1); (iii) Smallest third side 5: perimeter 19 cm (1); Largest third side 13: perimeter 27 cm (1); State the answers with units (1).	1+1+1+1+1

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using $90°$ or $360°$ instead of $180°$ for the angle sum of a triangle.

- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.
- In an isosceles triangle, taking the angles opposite the unequal side as equal.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 35	Exterior angle = sum of interior opposite angles.
2	**(a)** $\angle Q$	Use the idea of corresponding parts.
3	**(a)** $\angle Q$	The included angle lies between the two given sides.
4	**(d)** $35^\circ, 45^\circ, 100^\circ$	The three angles must add up to 180°.
5	**(d)** 50°	Equal angles and equal sides go together in an isosceles triangle.
6	**(a)** not necessarily congruent	AAA does not guarantee congruence.
7	**(b)** 8 cm	The third side lies between the difference and the sum of the other two.
8	**(b)** 36	Add the angles: $(3+k)x = 180$.
9	**(c)** 25	Exterior angle = sum of interior opposite angles.
10	**(b)** PQ	Use the idea of corresponding parts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$6x + 12 = 180$; $x = 28$ (1); Angles 74°, 72°, 34° (1).	1+1
12	Exterior angle $= \angle A + \angle B$; $5x - 6 = 3x + 50$, so $2x = 56$ (1); $x = 28$; $\angle ACD = 134^\circ$ (1).	1+1
13	Third side $> 8 - 11$ in absolute value $= 3$; Third side $< 8 + 11 = 19$ (1); Least whole number 4 cm; Greatest whole number 18 cm (1).	1+1
14	$6x + 30 = 180$; $x = 25$ (1); Angles 70°, 70°, 40° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = \angle C$ (equal sides); $(x + 30) + 2(2x - 25) = 180$, so $5x = 200$ (1); $x = 40$ (1); $\angle A = 70^\circ$, $\angle B = \angle C = 55^\circ$ (1).	1+1+1
16	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 15$ cm (1); Perimeter of PQR $= 8 + 10 + 15$ (1); $= 33$ cm (1).	1+1+1
17	$\angle A + \angle B = 130^\circ$; One part $= 130 \div 13 = 10^\circ$ (1); $\angle A = 60^\circ$, $\angle B = 70^\circ$ (1); $\angle C = 180^\circ - 130^\circ = 50^\circ$ (1).	1+1+1
	OR: $\angle B = \angle C$ (equal sides); $(x + 15) + 2(2x - 5) = 180$, so $5x = 175$ (1); $x = 35$ (1); $\angle A = 50^\circ$, $\angle B = \angle C = 65^\circ$ (1).	1+1+1
18	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 13$ cm (1); Perimeter of PQR $= 4 + 12 + 13$ (1); $= 29$ cm (1).	1+1+1
	OR: $\angle A + \angle B = 150^\circ$; One part $= 150 \div 5 = 30^\circ$ (1); $\angle A = 60^\circ$, $\angle B = 90^\circ$ (1); $\angle C = 180^\circ - 150^\circ = 30^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$4x + 40 = 180$, so $x = 35$ (1); Angles 65°, 25°, 90° (1); Classification: right-angled (1); Exterior angles 115°, 155°, 90° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: (i) Between 0 cm and 16 cm (1); (ii) Whole numbers 1 to 15: 15 values (1); (iii) Smallest third side 1: perimeter 17 cm (1); Largest third side 15: perimeter 31 cm (1); State the answers with units (1).	1+1+1+1+1
20	$4x + 44 = 180$, so $x = 34$ (1); Angles 74°, 26°, 80° (1); Classification: acute-angled (1); Exterior angles 106°, 154°, 100° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: (i) Between 9 cm and 21 cm (1); (ii) Whole numbers 10 to 20: 11 values (1); (iii) Smallest third side 10: perimeter 31 cm (1); Largest third side 20: perimeter 41 cm (1); State the answers with units (1).	1+1+1+1+1

Mistakes to avoid:

- Checking only one pair of sides in the triangle inequality; all three conditions must hold (the smallest pair is the critical one).
- Saying a triangle is possible when two sides add up exactly to the third; the sum must be *greater*.
- Using 90° or 360° instead of 180° for the angle sum of a triangle.
- Equating the exterior angle to the adjacent interior angle instead of the sum of the two interior opposite angles.

- In an isosceles triangle, taking the angles opposite the unequal side as equal.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **A Tale of Three Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** XY	The included angle lies between the two given sides.
2	**(b)** $20^\circ, 80^\circ, 80^\circ$	The three angles must add up to 180°.
3	**(c)** 45°	Equal angles and equal sides go together in an isosceles triangle.
4	**(a)** 12 cm	The third side lies between the difference and the sum of the other two.
5	**(a)** 50	Substitute x.
6	**(b)** 35	Exterior angle = sum of interior opposite angles.
7	**(a)** $\angle R$	Use the idea of corresponding parts.
8	**(d)** $15^\circ, 75^\circ, 90^\circ$	The three angles must add up to 180°.
9	**(c)** 3	Equal angles and equal sides go together in an isosceles triangle.
10	**(b)** 72	Substitute x.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Exterior angle $= \angle A + \angle B$; $5x - 8 = 3x + 42$, so $2x = 50$ (1); $x = 25$; $\angle ACD = 117^\circ$ (1).	1+1
12	Third side $> 5 - 5$ in absolute value $= 0$; Third side $< 5 + 5 = 10$ (1); Least whole number 1 cm; Greatest whole number 9 cm (1).	1+1
13	$6x + 6 = 180$; $x = 29$ (1); Angles 66°, 77°, 37° (1).	1+1
14	Exterior angle $= \angle A + \angle B$; $5x - 20 = 3x + 44$, so $2x = 64$ (1); $x = 32$; $\angle ACD = 140^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\angle B = \angle C$ (equal sides); $(x + 25) + 2(2x - 20) = 180$, so $5x = 195$ (1); $x = 39$ (1); $\angle A = 64^\circ$, $\angle B = \angle C = 58^\circ$ (1).	1+1+1
16	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 17$ cm (1); Perimeter of PQR $= 9 + 11 + 17$ (1); $= 37$ cm (1).	1+1+1
17	$\angle A + \angle B = 110^\circ$; One part $= 110 \div 11 = 10^\circ$ (1); $\angle A = 40^\circ$, $\angle B = 70^\circ$ (1); $\angle C = 180^\circ - 110^\circ = 70^\circ$ (1).	1+1+1
	OR: $\angle B = \angle C$ (equal sides); $(x + 5) + 2(2x - 20) = 180$, so $5x = 215$ (1); $x = 43$ (1); $\angle A = 48^\circ$, $\angle B = \angle C = 66^\circ$ (1).	1+1+1
18	Two sides and the included angle are equal, so SAS; $\triangle ABC \cong \triangle PQR$, hence $PR = AC = 12$ cm (1); Perimeter of PQR $= 7 + 8 + 12$ (1); $= 27$ cm (1).	1+1+1
	OR: $\angle A + \angle B = 100^\circ$; One part $= 100 \div 5 = 20^\circ$ (1); $\angle A = 20^\circ$, $\angle B = 80^\circ$ (1); $\angle C = 180^\circ - 100^\circ = 80^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$4x + 36 = 180$, so $x = 36$ (1); Angles 71°, 31°, 78° (1); Classification: acute-angled (1); Exterior angles 109°, 149°, 102° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: (i) Between 3 cm and 23 cm (1); (ii) Whole numbers 4 to 22: 19 values (1); (iii) Smallest third side 4: perimeter 27 cm (1); Largest third side 22: perimeter 45 cm (1); State the answers with units (1).	1+1+1+1+1
20	$4x + 16 = 180$, so $x = 41$ (1); Angles 63°, 29°, 88° (1); Classification: acute-angled (1); Exterior angles 117°, 151°, 92° (1); Their sum $= 360^\circ = 360^\circ$ (1).	1+1+1+1+1
	OR: (i) Between 5 cm and 13 cm (1); (ii) Whole numbers 6 to 12: 7 values (1); (iii) Smallest third side 6: perimeter 19 cm (1); Largest third side 12: perimeter 25 cm (1); State the answers with units (1).	1+1+1+1+1

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